

Bruce Cheng

bruce.zheng0814@gmail.com | [linkedin.com/in/bruce-lcz](https://www.linkedin.com/in/bruce-lcz) | github.com/bruce-lcz

EDUCATION

National Yunlin University of Science and Technology

Master of Science in Industrial Engineering and Management

Yunlin, Taiwan

2019 – 2021

National Yunlin University of Science and Technology

Bachelor of Science in Industrial Engineering and Management

Yunlin, Taiwan

2015 – 2019

EXPERIENCE

Information Engineer (Generative AI and LLM Projects)

PixArt Imaging Inc.

Aug. 2025 – Present

Hsinchu, Taiwan

- Building On-Premise Generative AI infrastructure meeting high security standards for IC Design industry.
- Automated contract comparison and patent translation using Vision-LLM OCR and multi-layer LLM pipelines.
- Constructed an offline GenAI collaboration platform driving 120B+ models for enterprise-wide productivity.
- Built the fully offline Auto-Minutes system, improving long-audio processing efficiency by 50%–70%.

AI Engineer / Project Leader

AUO Corporation - Advanced Manufacturing Center

Aug. 2021 – Aug. 2025

Taichung, Taiwan

- Led cross-functional teams to design AI architectures and optimize manufacturing processes.
- Increased cross-factory yield by 1.43% and reduced defects by 12.62% via AI parameter recommendation.
- Evolved intelligent assistant platform from RAG to a Multi-Agent system with Neo4j Knowledge Graph.
- Developed Predictive Maintenance system (PHM), reducing defect prediction error (RMSE) by 20%.
- Recognized as Level 3 Elite Talent in Smart Manufacturing; outcomes approved for internal Defensive Patent.

Research Intern

Academia Sinica - GIS Center

Feb. 2019 – Jul. 2019

Taipei, Taiwan

- Developed a Web system for address data sampling and normalization, solving unstructured data issues.
- Awarded Best Student Paper at TGIS Conference for the design of address attribute database modules.

PROJECTS

Legal Contract Comparison System | *Python, FastAPI, vLLM, Vision LLM, OCR*

2026

- Automated paper contract verification using Nanonets-OCR2-3B to extract text and filter signatures.
- Implemented diff-match-patch algorithm with Multiset Check to ensure zero-omission during restructuring.
- Reduced manual proofreading time from tens of minutes to seconds with dual-pane synchronized UI.

My Gym Tracker — Private AI Gym Coach | *React, TypeScript, Cloudflare Workers/D1*

2026

- Developed a serverless gym application using Neumorphism design and Cloudflare edge architecture.
- Integrated LLM for automatic muscle group mapping from plain-text exercise names.
- Designed prompt-engineered AI coaching to generate structured training recommendations from volume data.

Auto-Minutes Offline Transcription | *Whisper, Speaker Diarization, vLLM, Docker*

2025

- Chained Breeze ASR and pyannote.audio for accurate multi-speaker transcription in air-gapped environments.
- Optimized inference via VAD segmentation and GPU batch processing for long-audio recordings.
- Integrated local vLLM to perform automated punctuation correction and generate structured meeting summaries.

TECHNICAL SKILLS

Generative AI: On-Premise LLM (vLLM), RAG, LangGraph (Agents), Neo4j (Knowledge Graph), Multimodal LLM

Backend/Systems: Python (FastAPI/Django), REST API, Microservices, Cloudflare Workers/D1, Docker, AWS

Data Science: Time Series (LSTM), Explainable AI (LIME), Genetic Algorithms, Databricks, SQL/ETL

Frontend/Tools: Streamlit (AI Internal Tools), AI-Assisted Web Dev, Git, CI/CD, Linux Management